Cybersecurity Domain Project Report

Dissecting Packets: From HTTP to TLS

Packet Analysis using Wireshark-

Introduction

Network communication happens through the exchange of packets between devices. Understanding packet structure and protocol behavior is essential in cybersecurity, network troubleshooting, and traffic analysis.

This project focuses on analyzing real network traffic using Wireshark. The objective was to inspect **HTTP communication**, **DNS resolution**, and **TLS encrypted sessions** by capturing live packets and analyzing their protocol fields.

Objectives

The objectives of this project were:

- To capture live network packets.
- To analyze HTTP request-response communication.
- To observe DNS name resolution.
- To inspect TLS handshake and secure communication.
- To understand packet structure across multiple protocol layers.

Tools Used-

Tool	Purpose
Wireshark	Packet capture and protocol analysis
Web Browser	Generating HTTP/HTTPS traffic
Terminal / System DNS Resolver	Generating DNS requests

Experiment 1 — HTTP Packet Analysis

Filter Used

http

Procedure

1. Started packet capture in Wireshark.
2. Opened `http://example.com`.
3. Applied HTTP display filter.
4. Inspected HTTP GET packets.

Captured Information

Source IP Address

`2409:40d1:506:e37a:...`

Destination IP Address

`2606:4700:90c5:72db:...`

Source Port

`61139`

Destination Port

`80`

Request Method

`GET`

Host

`example.com`

Your capture shows:

- `GET / HTTP/1.1`
- Browser headers
- Host field
- User-Agent field

Packet Analysis

The client initiated an HTTP GET request to the server using port 80. Since HTTP is an unencrypted protocol, request headers and content were visible in plaintext.

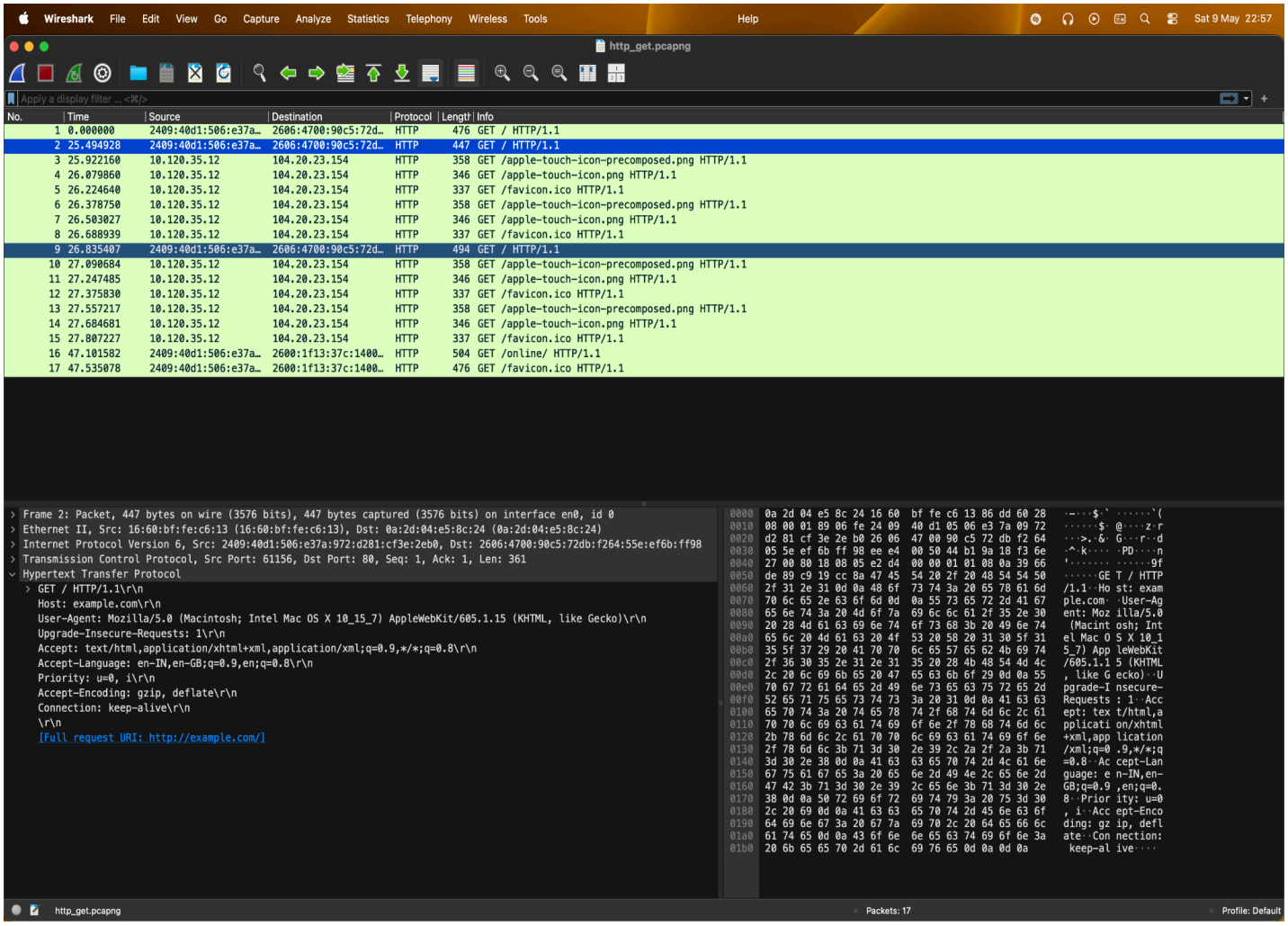
The captured packet revealed:

- Request URI
- Host name
- Browser information
- Accepted content types

Additional browser requests such as:

- `/favicon.ico`

were also observed.



Experiment 2 — DNS Packet Analysis

Filter Used

dns

Procedure

1. Generated DNS traffic by accessing domains.
2. Applied DNS display filter.
3. Inspected query and response packets.

Captured Information

Example Domain

`chatgpt.com`

Query Type

`A Record`

Source Port

`14393`

Destination Port

`53`

Resolved IP Addresses

The DNS server returned:

- `172.64.155.209`
- `104.18.32.47`

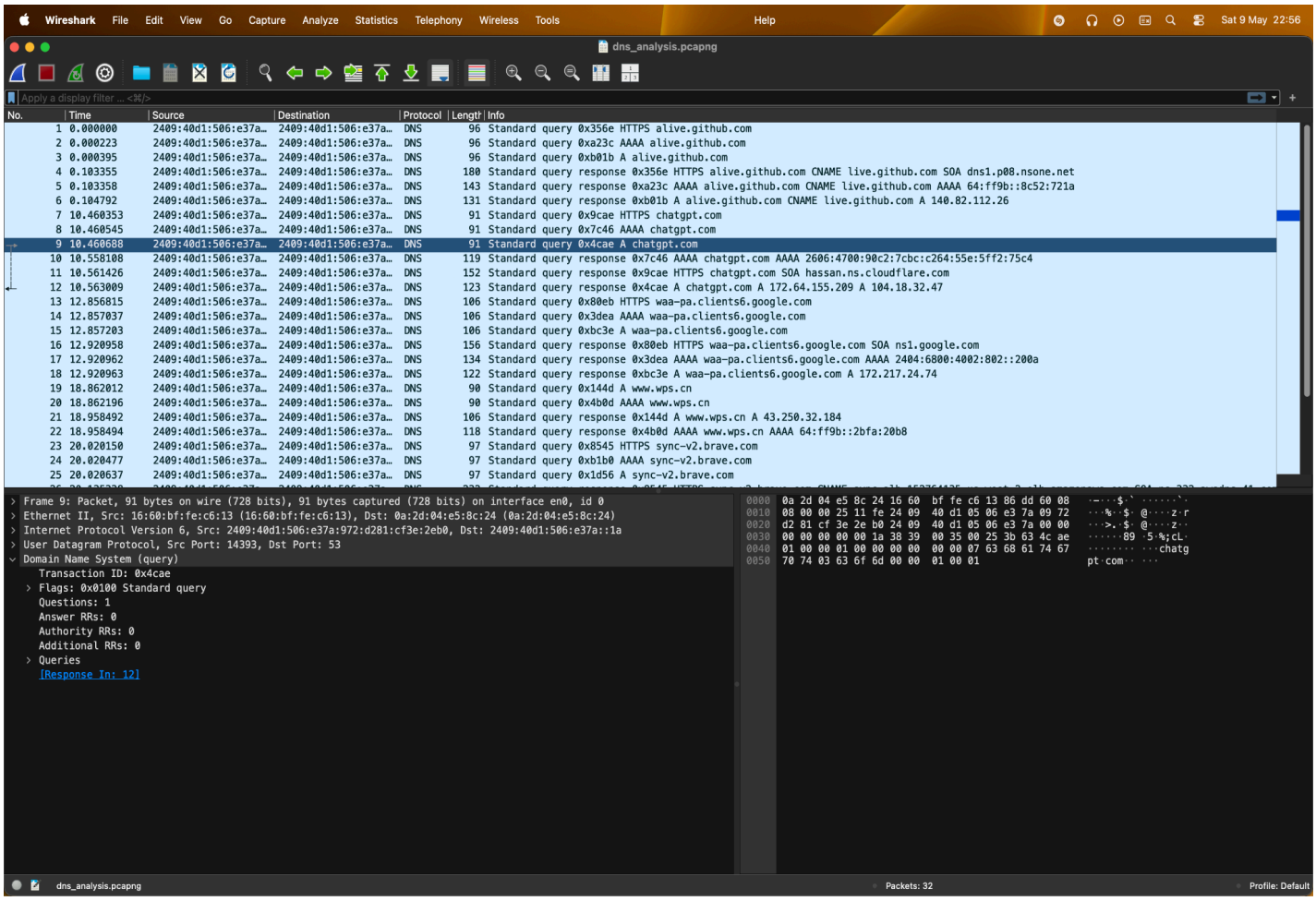
Packet Analysis

The client sent a DNS query requesting the IPv4 address of `chatgpt.com`.

The DNS server responded with valid IPv4 addresses. This process allows domain names to be translated into routable IP addresses.

Other DNS record types observed:

- A Record
- AAAA Record
- CNAME
- SOA



Experiment 3 — TLS Packet Analysis

Filter Used

tls

Procedure

1. Opened `https://example.com`
2. Applied TLS filter.
3. Inspected TLS handshake packets.

Captured Information

Destination Port

`443`

TLS Version

`TLS 1.3`

Server Name Indication

`example.com`

Captured handshake packets:

- Client Hello
- Server Hello

Packet Analysis

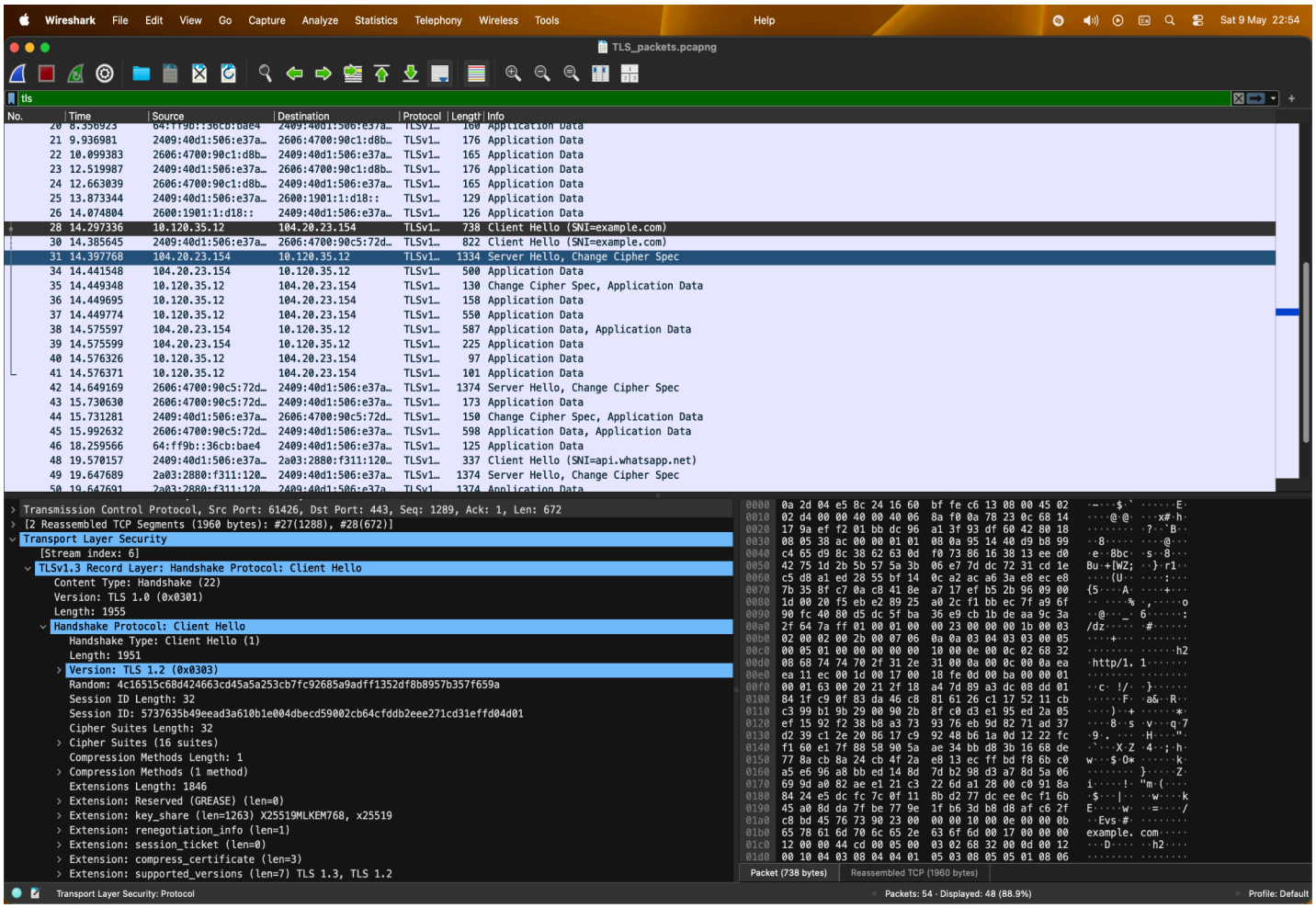
The client initiated a secure connection using TLS.

The **Client Hello** packet contained:

- Supported TLS versions
- Cipher suites
- Session ID
- Random values
- Server Name Indication (SNI)

The server responded with **Server Hello**, selecting encryption parameters.

After successful negotiation, encrypted communication began.



Comparison of Protocols-

Protocol	Port	Purpose	Security
HTTP	80	Web communication	Not encrypted
DNS	53	Domain resolution	Not encrypted (traditional DNS)
TLS	443	Secure communication	Encrypted

Conclusion

This project provided practical experience in packet analysis using Wireshark. By analyzing HTTP, DNS, and TLS traffic, it was possible to understand how network communication occurs across different layers.

The project improved understanding of:

- Packet structure
- Protocol communication
- DNS resolution
- Secure TLS negotiation
- Network security fundamentals

Packet Capture Files

The captured packet logs used in this project can be accessed using the following link:

Wireshark Capture Files:

Wireshark Project Logs

Files included:

HTTP packet capture

DNS packet capture

TLS packet capture